



Characteristics and controlling factors of soil micro-climate in a subtropical re-vegetation catchment: Insights from continuous moisture and temperature observations

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Abstract

Background and aims Global increases in vegetation greenness are largely attributed to ecological restoration efforts. Soil micro-climate (moisture and temperature) serves as the foundation for vegetation recovery and constitutes a critical component of ecosystem services. However, in subtropical humid mountainous regions with complex terrain, the spatial patterns and driving factors of soil micro-climate remain poorly understood.

Methods From 2020 to 2021, high-resolution (5-min intervals) soil moisture and temperature data were continuously collected to characterize soil micro-climate

across four topographic positions (lower valley, upper valley, south-facing slope, and north-facing slope) in a re-vegetated catchment in China's Three Gorges Reservoir Area.

Results Soil moisture exhibited strong spatial heterogeneity, being higher in valley than slopes. The lower valley had the highest soil moisture content ($0.36 \text{ cm}^3 \text{ cm}^{-3}$), while north-facing slope had the lowest ($0.26 \text{ cm}^3 \text{ cm}^{-3}$). Soil temperature within the profile varied seasonally, increasing by 5.0 to 9.2 °C during the rainy season. Additionally, seasonal fluctuations of temperature were smaller in deeper soil layers, highlighting the thermal buffering capacity of subsurface soils. Mantel test revealed that slope position exerted its influence on the soil micro-climate through differences in vegetation canopy closure, root depth and soil texture. During the rainy season, air temperature (41.1% contribution) and relative humidity (46.1% contribution) were the primary factors influencing soil micro-climate. In contrast, during the non-rainy season, air temperature became the most significant factor, with an average contribution exceeding 55%.

Conclusions Our findings highlight the spatiotemporal variability of soil micro-climate and its key controlling factors in subtropical mountainous ecosystems. These insights can provide guidance for developing more effective and targeted vegetation restoration strategies, including the selection of appropriate species and the positioning of restoration sites. Moreover, further validation is needed to refine these applications for practical use.

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Introduction

Soil constitutes the fundamental medium for supporting terrestrial ecosystems, facilitating their existence, evolution, and long-term development (Robinson et al. 2019). The patterns of soil moisture and temperature, which constitute the soil micro-climate, play a crucial role in influencing various processes and interactions within the near-surface earth system (Melillo et al. 2002; Seneviratne et al. 2010). Variations in soil moisture and temperature conditions are pivotal in regulating biological processes across ecosystems (Miguez-Macho and Fan 2021). Landscape heterogeneity at localized scale can buffer the impacts of regional climatic patterns, resulting in soil micro-climates that deviate significantly from broader environmental conditions (Lenoir et al. 2017; Gilbert et al. 2022). Numerous studies have demonstrated that these micro-climatic variations in soil could notably shape biogeochemical dynamics, water cycling, and vegetation growth at the surface level (Fan et al. 2020; Bassiouni et al. 2023).

Soil moisture is widely recognized as a key factor linking hydrological, biological, and atmospheric systems (Albergel et al. 2012; Fang et al. 2016). It plays a decisive role in modulating climate through its influence on energy exchange and hydrological partitioning, such as infiltration, surface runoff, and rainfall response (Guo and Lin 2018; Singh et al. 2021). Similarly, soil temperature governs energy dynamics at the ground surface and gas fluxes between the land and atmosphere, affecting both hydrological cycles and nutrient transformations (Seyfried et al. 2016; Yu et al. 2017). Together, soil moisture and temperature are central to processes such as runoff initiation, soil evaporation, and vegetation development by influencing plant water-use strategies (Xu et al. 2025). Understanding soil micro-climate dynamics is crucial for advancing ecohydrological research and predicting responses to climatic changes (Wundram et al. 2010; Trnka et al. 2013). However, the spatial and temporal variability of soil micro-climates, particularly in complex terrains, remains insufficiently understood (Fekete et al. 2016; Leuthold et al. 2021).

Investigating soil moisture and temperature variations presents an opportunity to gain deeper insights into land surface system behaviors.

Given the ecological significance of soil climatic conditions, there is growing interest in quantifying and simulating their spatial-temporal patterns (Solomon et al. 2007). Sensor-based monitoring networks for soil moisture and temperature have provided new avenues for characterizing soil micro-climates (Ritsema et al. 2009). These high-resolution monitoring networks have been implemented in diverse ecosystems, ranging from agricultural landscapes (De Lanoy et al. 2006; Li et al. 2013; Zhu et al. 2014) to forested watersheds (Bogena et al. 2010; Fan et al. 2020) and karst depressions (Zhang et al. 2023). These studies have enhanced our understanding of land surface system behaviors (Dymond et al. 2021). However, in steep and mountainous terrains, soil micro-climates exhibit substantial heterogeneity due to complex topographic influences on hydrology and energy fluxes (Penna et al. 2013; Wilson et al. 2020). Despite this, most existing research remains localized, focusing on individual soil profiles or specific positions, limiting our understanding of broader landscape-scale patterns and processes (Yi et al. 2019; Lu et al. 2024).

Topographical elements are recognized as dominant factors influencing soil moisture and temperature regimes, from the hillslopes to the catchment scale (Western et al. 2004; Gutiérrez-Jurado and Vivoni 2013). Some studies have indicated that slope aspect significantly alters the micro-climate, with north-facing slopes (in the Northern Hemisphere) generally being cooler and moister, while south-facing slopes receive more solar radiation, resulting in warmer and drier conditions (Burnett et al. 2008; Geroy et al. 2011). However, Reid (1973) observed that in energy-limited systems, higher radiation would not always correlate with reduced soil moisture. Moreover, the positions along a slope (upper, middle, or lower) can further modify soil moisture and temperature, with downslopes typically retaining more moisture and exhibiting cooler profiles (Zhu et al. 2014; Jin et al. 2019). These topographic and climatic gradients contribute to complex spatiotemporal variations in soil micro-climate, underscoring the need for long-term, continuous observations.

Soil moisture and temperature conditions provide the foundation for vegetation growth. The spatial heterogeneity of the soil micro-climate determines

the effectiveness of vegetation restoration (Jin et al. 2019; Xu et al. 2025). While the soil micro-climate influences the vegetation restoration process, it is also regulated by vegetation structural characteristics (Brantley et al. 2017; López-Vicente et al. 2023). Through processes such as transpiration and litter accumulation, vegetation covers can alter soil moisture and temperature conditions, thus influencing ecosystem functions (Song et al. 2020; Mao et al. 2024). Furthermore, soil properties impact the soil micro-climate by affecting hydraulic and thermal characteristics (Alaoui et al. 2011; Zhang et al. 2023). Studies have shown that the soil micro-climate may experience significant seasonal and interannual variations in response to meteorological conditions (Ghannam et al. 2016; Fan et al. 2020). In vegetation restoration areas with complex topography, the integrated effects of terrain, vegetation, and climate on the soil micro-climate remain underexplored and lack systematic research (Fan et al. 2020; Tanarhte et al. 2024).

In the Three Gorges Reservoir area of China, ecological restoration faces multiple challenges, including rugged terrain, heavy rainfall, and significant changes in land cover (Lu et al. 2023). In this context, there is an urgent need to better understand the spatiotemporal variability of soil micro-climate to guide effective vegetation restoration strategies (Xu et al. 2024) and to promote the practical application in species selection, site prioritization, and soil and water conservation strategies (Yi et al. 2019; Lu et al. 2024). To address this gap, this study employed high-resolution monitoring of soil moisture and temperature in a re-vegetation catchment within the Dalaoling Nature Reserve to address three core questions: (1) What are the spatiotemporal distribution patterns of soil micro-climate at the catchment scale? (2) What are the primary controlling factors of the micro-climate in subtropical mountainous regions? (3) How can high-resolution monitoring data guide vegetation restoration and management strategies? We hypothesize that: (1) The topographic position within a catchment significantly influences the spatiotemporal distribution of soil moisture and temperature, with a more pronounced impact on soil moisture due to scale limitations. (2) Vegetation characteristics and climatic conditions regulate the distribution pattern of soil micro-climate, exhibiting seasonal variations. The results of this study are expected to contribute valuable insights for guiding soil conservation strategies,

improving vegetation management, and enhancing climate resilience in forest ecosystems of the region.

Materials and methods

Study area

This study was conducted in the Dalaoling Nature Reserve (30°00'13"–31°28'30"N, 110°51'8"–111°39'30"E), located in the Three Gorges Reservoir Area with an average elevation of 1300 m (Fig. 1a). The area is characteristic of a typical mid-latitude mountainous region at the northern edge of China's central subtropical zone, characterized by a subtropical monsoon climate. The mean annual temperature is 16.7 °C, with the highest and lowest monthly averages occurring in July and January, respectively. Annual precipitation ranges from 1000 to 1340 mm, and the frost-free period lasts approximately 280 days. Precipitation is concentrated primarily from May to September, with over 50% of the precipitation contributed by heavy rain and rainstorms. Rainfall intensity peaks during July and August, with maximum rates reaching up to 48 mm/h (Liu et al. 2023). The terrain exhibits distinct topographic variation, with a southwest-high and northeast-low orientation, resulting in significant relief. Based on the soil profile observations, the soil type in the study catchment is classified as boralf according to the USDA soil taxonomy. Due to human disturbance, the native vegetation has been significantly degraded, with scarce remnants of primary evergreen broadleaf forests. The current vegetation is predominantly secondary or plantation forests (Lu et al. 2023).

Field surveys were conducted to select a representative catchment as our study area (Fig. 1b, c). We consider this catchment to be typical because it is covered by restored forests representative of the region and contains characteristic topographic features such as valley and side-slopes. The catchment spans an elevation ranging from 1330 to 1400 m, with a slope length of ~170 m from base to ridge and gradients of 15–25°. The soil layer thickness varies from 0.8 to 1.6 m. The catchment has undergone vegetation restoration, involving both artificial and natural regeneration. The planted forests primarily consist of evergreen coniferous species that are over two decades old, whereas the naturally regenerated sections feature a mix of secondary deciduous trees and shrubs

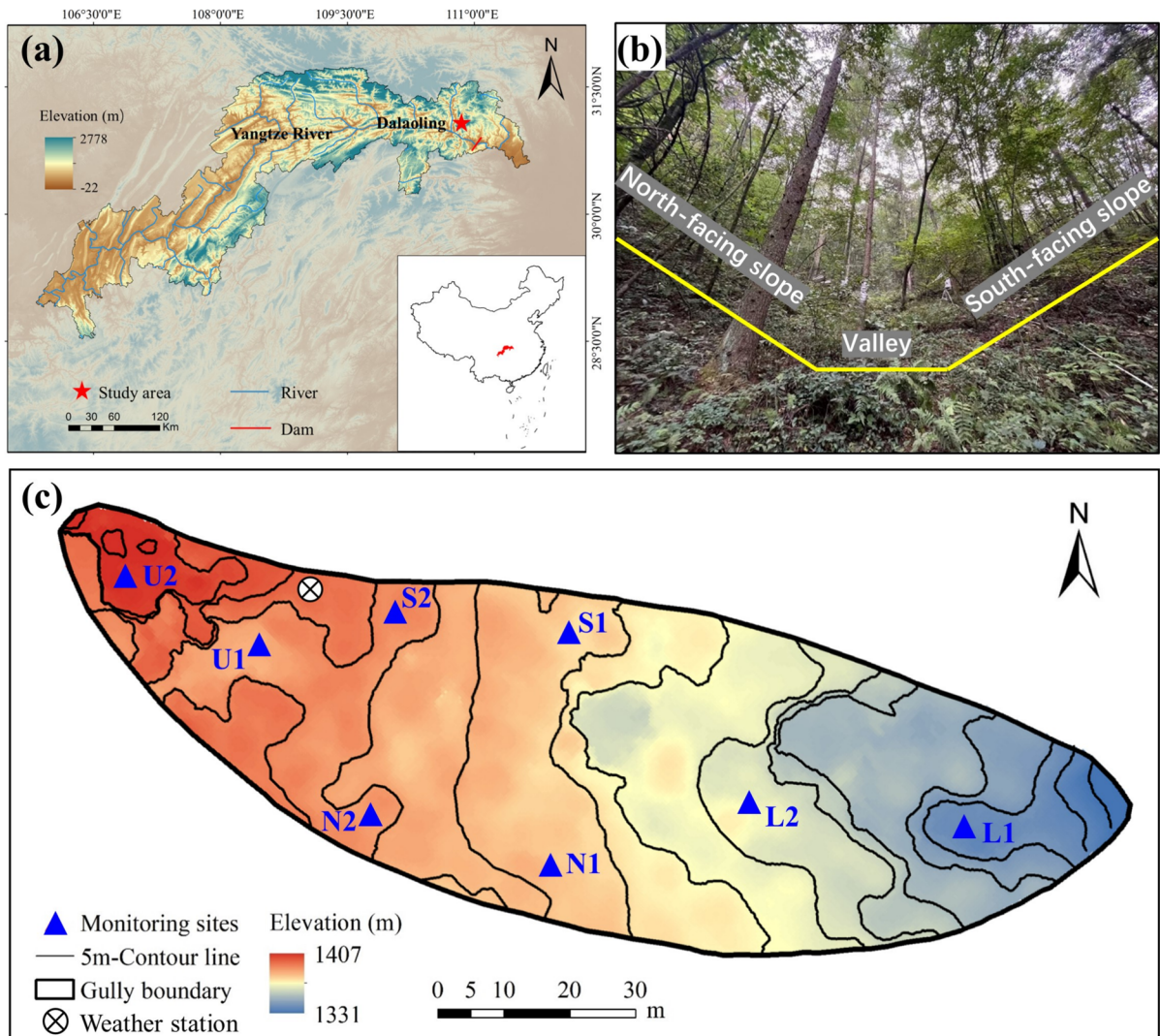


Fig. 1 The location of Dalaoling Nature Reserve in the Three Gorges Reservoir Area (a), landscape of the re-vegetation catchment (b), and the spatial distribution of eight monitoring

sites in the catchment (c). L1 and L2, lower valley 1 and 2; U1 and U2, upper valley 1 and 2; S1 and S2, south-facing slope 1 and 2; N1 and N2, north-facing slope 1 and 2

aged 9 to 12 years, growing on land previously subject to deforestation (Xu et al. 2024). Based on our previous investigation, the entire catchment is covered with planted *Cunninghamia lanceolata* (Lamb.) Hook., which is the dominant vegetation type at each monitoring site. In addition, small amounts of *Bambusoideae* are present at the north-facing slope and upper valley. The south-facing slope and lower valley feature some *Quercus* species. Herbaceous plants are distributed across all monitoring sites, with the main species including *Miscanthus sinensis*, *Rubus corchorifolius* L.f., and *Osmunda japonica*.

Experiment design

Based on field surveys, the catchment is composed of three distinct topographic units: valley, south-facing slope, and north-facing slope (Fig. 1b, c). To gain a comprehensive understanding of the micro-climate within the catchment, eight monitoring sites were established (Table 1): lower valley 1 and 2 (L1, L2); upper valley 1 and 2 (U1, U2); south-facing slope 1 and 2 (S1, S2); north-facing slope 1 and 2 (N1, N2). Among these sites, L1, U1, S1, and N1 were located

Table 1 Basic characteristics of eight monitoring sites in the selected forested catchment

Topographic positions	Monitoring sites	Slope gradient (°)	Slope aspect (°)	Altitude (m)	Soil depth (cm)	Mean canopy closure (%)	Root depth (cm)
Lower valley	L1	17	-	1335	160	91.34 ± 4.35	102.3
	L2	26	-	1346	140	87.68 ± 3.26	100.5
Upper valley	U1	21	-	1364	155	82.97 ± 5.87	95.2
	U2	18	-	1385	100	75.72 ± 2.89	88.6
South-facing slope	S1	23	180	1340	140	88.28 ± 4.66	96.3
	S2	21	180	1365	110	77.68 ± 3.16	84.2
North-facing slope	N1	19	0	1342	150	87.09 ± 5.67	98.5
	N2	20	0	1355	145	80.03 ± 2.34	92.6

at relatively lower positions within each respective topographic unit, while L2, U2, S2, and N2 were positioned at higher elevations.

An automatic weather station was installed in an open area of the study catchment to monitor key meteorological parameters, including rainfall, air temperature, relative humidity, solar radiation and wind velocity (Fig. 1c). The weather station was assembled by Yangling Zhiming Ecological Technology Inc. of China. Rainfall was measured using a tipping-bucket rain gauge (0.2 mm/tip; Jiangsu Naiweh Corporation Inc., China), recording at 30-min intervals. Air temperature and relative humidity were measured with an HMP50 temperature-humidity probe (Vaisala Inc., Finland). Solar radiation and wind velocity were monitored using a LI-200 radiation sensor and an RM03001 anemometer (Yangling Zhiming Ecological Technology Inc., China), respectively. Data were

recorded by a CR300 data logger (Campbell Scientific Inc., Logan, Utah, USA).

Before the installation of the monitoring system, soil profiles at each monitoring site were excavated and thoroughly described. The soil structure and root length of all eight profiles were recorded. In addition, environmental factors such as elevation, slope gradient, and canopy closure were documented for each site. At each location, a total of 400 undisturbed soil samples (8 sites × 5 soil depths × 10 replicates) were collected using standard cutting rings (100 cm³). Concurrently, approximately 500 g of disturbed soil samples were collected from each soil layer for analysis. These samples were used to determine key soil properties (Table 2), including soil texture (sand, silt, and clay content), organic matter content (OMC, g/kg), saturated hydraulic conductivity (K_s , cm/h), porosity (P, %), and bulk density (BD, g/cm³). The

Table 2 Average soil properties of eight monitoring sites in the study catchment

Monitoring sites	SA (%)	SI (%)	CL (%)	BD (g cm ⁻³)	P (%)	K_s (cm h ⁻¹)	OMC (g kg ⁻¹)
L1	39.85 ± 2.46c	37.59 ± 2.01ab	22.56 ± 3.10a	0.92 ± 0.07a	46.78 ± 13.43ab	1.92 ± 0.74b	58.17 ± 44.45a
L2	38.90 ± 4.79c	40.08 ± 4.56ab	21.02 ± 2.70ab	1.04 ± 0.26a	50.40 ± 25.09ab	1.45 ± 1.26b	47.57 ± 54.80a
U1	36.22 ± 3.73c	43.69 ± 3.90a	20.09 ± 2.92ab	0.89 ± 0.22a	53.01 ± 6.65ab	1.56 ± 0.96b	31.40 ± 16.69a
U2	50.72 ± 15.48b	34.10 ± 14.25bc	15.19 ± 2.10 cd	1.10 ± 0.25a	42.93 ± 15.65b	3.52 ± 0.75a	40.76 ± 19.54a
S1	51.86 ± 8.11b	27.32 ± 6.19 cd	20.82 ± 9.02ab	1.01 ± 0.21a	62.77 ± 9.41a	1.53 ± 0.82b	26.02 ± 29.73a
S2	62.04 ± 5.85a	26.07 ± 3.38d	11.89 ± 2.55d	1.03 ± 0.14a	54.50 ± 13.02ab	2.17 ± 0.27b	25.11 ± 16.78a
N1	43.77 ± 2.46bc	40.88 ± 2.05ab	15.35 ± 2.12dc	0.98 ± 0.25a	62.05 ± 7.82a	2.09 ± 0.52b	35.80 ± 32.43a
N2	42.97 ± 7.15bc	40.08 ± 4.65ab	16.95 ± 3.10bc	1.20 ± 0.34a	62.49 ± 4.01a	1.31 ± 0.99b	30.05 ± 32.07a

L1 and L2, lower valley 1 and 2; U1 and U2, upper valley 1 and 2; S1 and S2, south-facing slope 1 and 2; N1 and N2, north-facing slope 1 and 2. SA, sand content; SI, silt content; CL, clay content; BD, bulk density; P, porosity; K_s , saturated hydraulic conductivity; OMC, organic matter content. The different letters represent significant differences at the 0.05 level between eight monitoring sites

soil particle composition was measured using the pipette method; porosity was determined using the water suction method (Li and Shao 2006); bulk density was measured through oven-drying; organic matter content was analyzed by the potassium dichromate oxidation method (Walkley and Black 1934); and saturated hydraulic conductivity was assessed with the constant head method, based on Darcy's law (Ferrer-Julia et al. 2021).

Soil moisture and temperature monitoring

To quantify real-time changes in soil micro-climate, we installed automated soil moisture-temperature monitoring systems (5TM sensors with ± 0.08 vol% precision and ± 3 vol% accuracy; METER Group, Inc., Pullman, WA, USA) at eight monitoring sites distributed across different topographic positions within the catchment. Previous investigations indicated that soil moisture showed considerable variation above a depth of 80 cm, whereas deeper layers tended to exhibit more stable conditions (Lu et al. 2024; Xu et al. 2024). In addition, the selected depths correspond to key transition zones between the O horizon and the C horizon. Therefore, at each monitoring site, five probes were installed at different soil depths (0–10 cm, 10–20 cm, 20–40 cm, 40–60 cm, and 60–80 cm), corresponding to the O, A, B, BC, and C horizons, respectively.

To ensure the reliability of the monitoring data, three quality control procedures were applied (Dorigo et al. 2013; Wickenkamp et al. 2016; Gao et al. 2020). First, to reduce the effect of soil disturbance on natural water flow, the sensors were installed on the upslope side within the soil pit. After installation, the soil was carefully refilled and compacted to restore its original bulk density. Second, in order to avoid the influence of moisture equilibration processes following disturbance, data collected during the first six months were excluded from the analysis. Third, inconsistent or erratic measurements resulting from technical issues such as sensor malfunction or low battery levels were identified through visual inspection and subsequently removed from the dataset. Soil moisture and temperature data were recorded at 5-min intervals using EM-50 data loggers (Decagon Devices, Inc., Pullman, WA, USA). We obtained high-quality monitoring data spanning from April 2020 to April

2021. In this study, the raw data were averaged to produce soil moisture and temperature readings at 30-min intervals.

The daily average soil moisture (θ , $\text{cm}^3 \text{ cm}^{-3}$) and temperature (T , $^{\circ}\text{C}$) for monitoring site i (θ_i , $\text{cm}^3 \text{ cm}^{-3}$; T_i , $^{\circ}\text{C}$) was calculated as the depth-weighted daily average soil moisture (θ_n , $\text{cm}^3 \text{ cm}^{-3}$) and depth-weighted daily average soil temperature (T_n , $^{\circ}\text{C}$) of the five soil layers (0–10, 10–20, 20–40, 40–60, and 60–80 cm) at site i (Fan et al. 2020):

$$\theta_i = \sum_{n=1}^5 d_n \theta_n / \sum_{n=1}^5 d_n \quad (1)$$

$$T_i = \sum_{n=1}^5 d_n T_n / \sum_{n=1}^5 d_n \quad (2)$$

where d_n (cm) is the depth of a soil layer, n indicates the n^{th} soil layer (with a value of 1, 2, 3, 4 and 5, referring to soil layers of 0–10, 10–20, 20–40, 40–60, and 60–80 cm).

Statistical analysis

To facilitate comparison at both annual and seasonal scales, daily averages of soil moisture and temperature were calculated from measurements recorded at five-minute intervals. Variations in these parameters across eight distinct monitoring locations, encompassing multiple depths and time periods, were statistically assessed using one-way analysis of variance (ANOVA). To explore the relationships between soil micro-climate and influencing factors such as vegetation characteristics, topographic features, and soil physicochemical attributes, Mantel test analysis was performed. All statistical computations were performed at a 95% confidence level ($\alpha=0.05$) using SPSS version 22.0 (SPSS Inc., USA). Data visualization and plotting were carried out using Origin 2022 (OriginLab, Northampton, MA, USA).

To further evaluate how meteorological variables (including precipitation, air temperature, relative humidity, solar radiation, and wind velocity) affect soil micro-climate over different temporal scales, a boosted regression tree (BRT) modeling framework was employed. Known for its capacity to handle both regression and classification tasks, BRT is particularly effective in capturing nonlinear relationships, dealing with multicollinearity, managing incomplete datasets, and quantifying the relative influence of

predictor variables (Rodeghiero et al. 2018; Sanaei et al. 2019). According to the recommended optimal settings for empirical ecological studies (Elith et al. 2008; Zhang et al. 2012; Lin et al. 2016; Yuan et al. 2018; Sanaei et al. 2019), we fitted BRT models using the combinations of the following parameters: a tree complexity of 5, learning rates of 0.001, bag fractions of 0.60, and tenfold cross-validations. All modeling procedures were conducted using R 4.3.3 (R Core Team 2024), incorporating the *vegan* (v2.6–8; Oksanen et al. 2024), the *dismo* (v1.3–14; Hijmans et al. 2023), the *gbm* (v2.2.2; Ridgeway and Developers 2024) packages.

Results

Spatiotemporal distribution of soil moisture and temperature

There was significant spatial variability in daily soil moisture across the monitoring sites (Fig. 2 and Fig. 3a), indicating a strong relationship between soil moisture and the topographic position. Overall, soil moisture content (SMC) within the catchment varied between 0.13 and 0.47 cm³ cm⁻³. A decreasing trend in average SMC from the bottom to the top of the valley was observed, and the south-facing slope consistently exhibited higher moisture levels compared to the north-facing slope. The coefficient of variation (CV) for soil moisture ranged from 8.7% to 28.9% across all sites. The lower valley had a higher CV (14.9%) than the upper valley (10.7%). The north-facing slope displayed the highest variability (CV = 22.9%), which was significantly greater than that observed on the south-facing slope (CV = 10.1%).

At most valley sites (L1, L2, U1), soil moisture increased with soil depth. However, on both the south-facing and north-facing slopes, particularly at sites S2, N1, and N2, soil moisture generally decreased with depth, and this pattern was particularly pronounced on the north-facing slope. In addition, daily soil moisture exhibited interannual temporal variability, mainly influenced by climatic differences among seasons, such as temperature and rainfall conditions.

In contrast, daily soil temperature exhibited relatively low spatial variability and was closely correlated with air temperature, showing pronounced

seasonal variation (Fig. 2 and Fig. 3b). Soil temperature across all monitoring sites was higher during the rainy season (April to September) and lower during the non-rainy season (October to March of the following year). Overall, soil temperature across the catchment ranged from 1.4 °C to 22.2 °C. Within the valley, average soil temperature increased from the bottom to the top (11.1 °C to 12.2 °C). On the slopes, S1 had a higher mean soil temperature (12.0 °C) than N1 (11.6 °C), while S2 and N2 showed similar values. The CV for soil temperature ranged between 39.5% and 45.4%. Within the valley, the CV gradually decreased from L1 to U2, and was higher on the south-facing slope compared to the north-facing slope.

Soil micro-climate in different topographic positions and seasons

The spatial distribution of soil moisture across different topographic positions was influenced by seasonality, and this seasonal difference was more pronounced in the shallower soil layers (Fig. 4). Additionally, significant differences in soil moisture were observed among topographic positions. The valley site group (L; U) exhibited higher SMC across all soil layers compared to the side-slope site group (S; N). In the shallower layers (0–10 and 10–20 cm), the differences in soil moisture among topographic positions were relatively small (Fig. 4a, b), while in the deeper layers (20–40, 40–60, and 60–80 cm), more pronounced spatial variations were observed (Fig. 4c–e). In particular, at depths of 20–40 cm and 40–60 cm, the lower valley showed significantly higher soil moisture compared to other monitoring sites. However, at the 60–80 cm depth, soil moisture in the lower valley decreased noticeably. Similarly, the north-facing slope also exhibited a clear decrease in soil moisture in the deeper layers. Moreover, seasonal fluctuations in soil moisture were more pronounced during the rainy season, especially in the shallower soil layers, where the changes were most evident.

Soil temperature exhibited clear seasonal spatial variation, with significantly ($p < 0.05$) higher mean values during the rainy season compared to the non-rainy season (Fig. 5). The largest temperature difference occurred in the 0–10 cm layer (Fig. 5a), and the seasonal temperature difference gradually decreased with increasing soil depth. Furthermore, the spatial

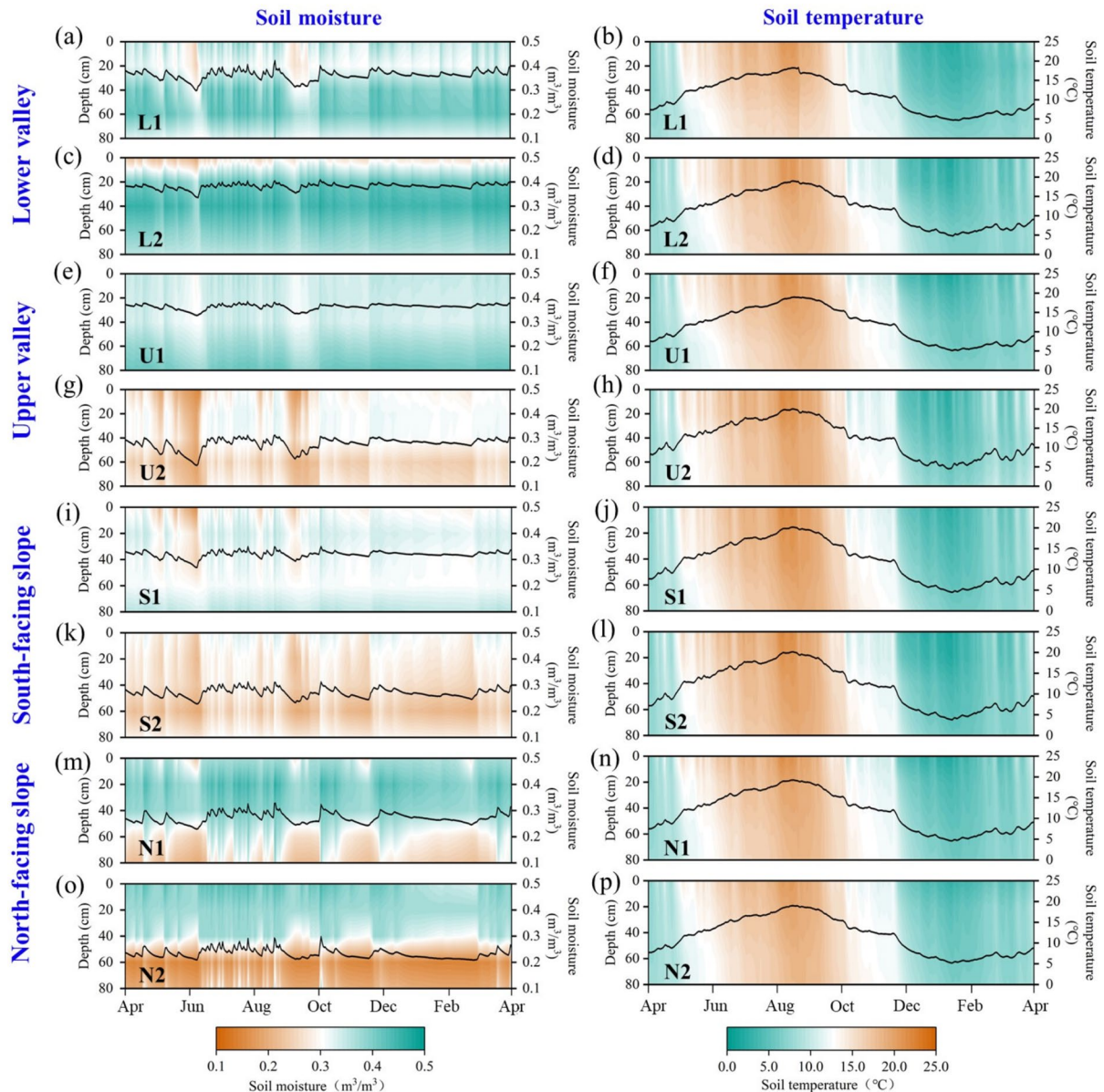


Fig. 2 Dynamics of soil moisture and temperature at different soil depths over the study period (April 1, 2020 to April 1, 2021) at eight monitoring sites within the study catchment

distribution of soil temperature across different topographic positions varied between the rainy and non-rainy seasons. During the rainy season, soil temperature at the south-facing slope site was higher than at other locations. In contrast, during the non-rainy season, soil temperature at the upper valley site was higher. Overall, at most soil depths, soil temperature was lower at the north-facing slope and lower valley sites compared to the south-facing slope and upper valley sites.

Controlling factors on soil micro-climate

Vegetation, topography, and soil physicochemical properties

Key vegetation, topographic, and soil characteristics exhibited varying degrees of influence on soil moisture and temperature (Fig. 6). Soil moisture was significantly positively correlated with mean canopy

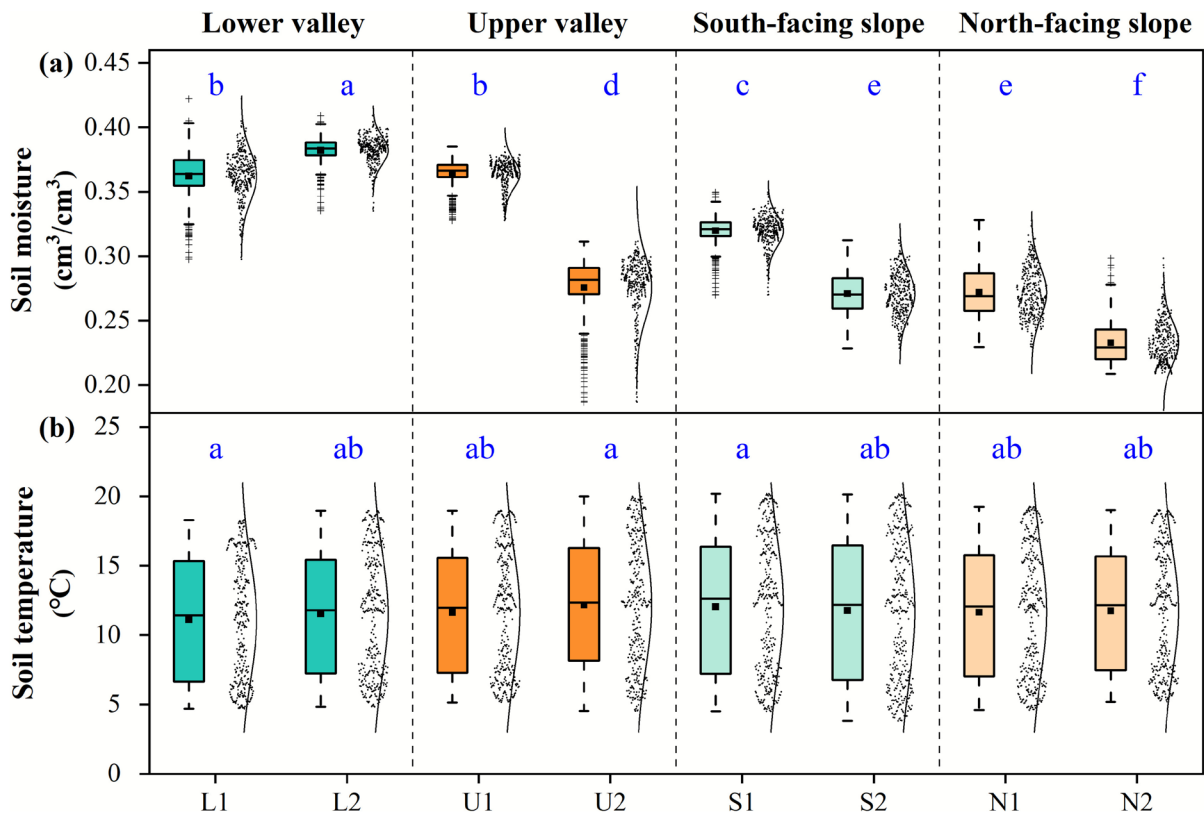


Fig. 3 Daily mean soil moisture (a) and soil temperature (b) across the eight monitoring sites throughout the study period. Letters above the error bars denote statistically significant dif-

ferences ($p < 0.05$) in mean soil moisture and temperature values between the monitoring sites

closure, root depth, and clay content ($p < 0.01$), while it was significantly negatively correlated with sand content and bulk density ($p < 0.01$). The influence of topographic factors on soil moisture was relatively weak. In contrast, soil temperature was significantly negatively correlated with mean canopy closure and root depth ($p < 0.01$), and positively correlated with altitude ($p < 0.01$). Additionally, soil temperature was significantly influenced by soil texture, bulk density, and organic matter content ($p < 0.05$). Notably, soil moisture and temperature exhibited a significant negative correlation ($p < 0.05$).

Meteorological variables

Meteorological variables exerted distinct controls on soil moisture and temperature, as evidenced by the

types of controlling variables and their performance across seasons (Fig. 7). Soil moisture was primarily affected by air temperature and relative humidity, with their combined relative contribution exceeding 65% in both seasons (Fig. 7a, b). Specifically, soil moisture was mainly controlled by air temperature during the rainy season, whereas it was predominantly influenced by relative humidity during the non-rainy season. In contrast, soil temperature was chiefly influenced by air temperature, relative humidity, and solar radiation (Fig. 7c, d). The relative contribution of air temperature to soil temperature was significantly higher than that of other variables, exceeding 50% in both seasons. Notably, wind velocity and precipitation had relatively minor effects on both soil moisture and temperature, with the contribution of wind velocity remaining below 10% across all seasons.

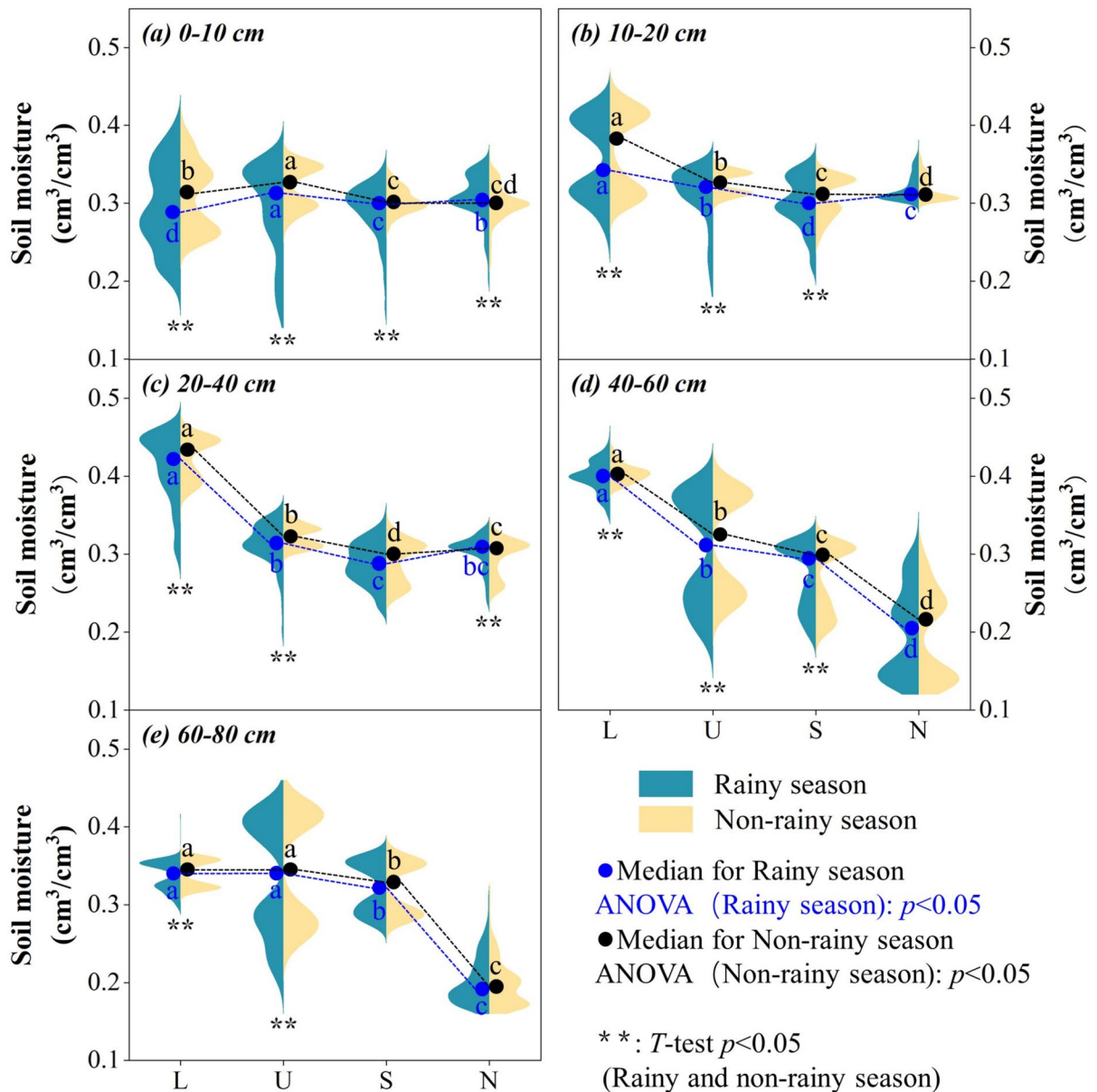


Fig. 4 A comparison of the variability of soil moisture between the rainy and non-rainy seasons in five soil layers. The left half of the violin plot shows the probability distribution of daily soil moisture in rainy season, while the right half depicts soil moisture in non-rainy season. Blue circles depict the median soil moisture in rainy season, while black circles depict non-rainy season. Different letters indicate significant

($p < 0.05$) differences in average soil moisture among site groups in each season. ** indicates a significant difference ($p < 0.05$) in average soil moisture between two seasons in t -tests. L, lower valley site group; U, upper valley site group; S, south-facing slope site group; N, north-facing slope site group. Each site group includes two monitoring sites

Discussion

Effects of topographic positions on soil micro-climate in different seasons

Our findings indicated that both soil moisture and temperature exhibited distinct spatial heterogeneity influenced by topographic position within the catchment (Figs. 2–5). Soil moisture tended to accumulate

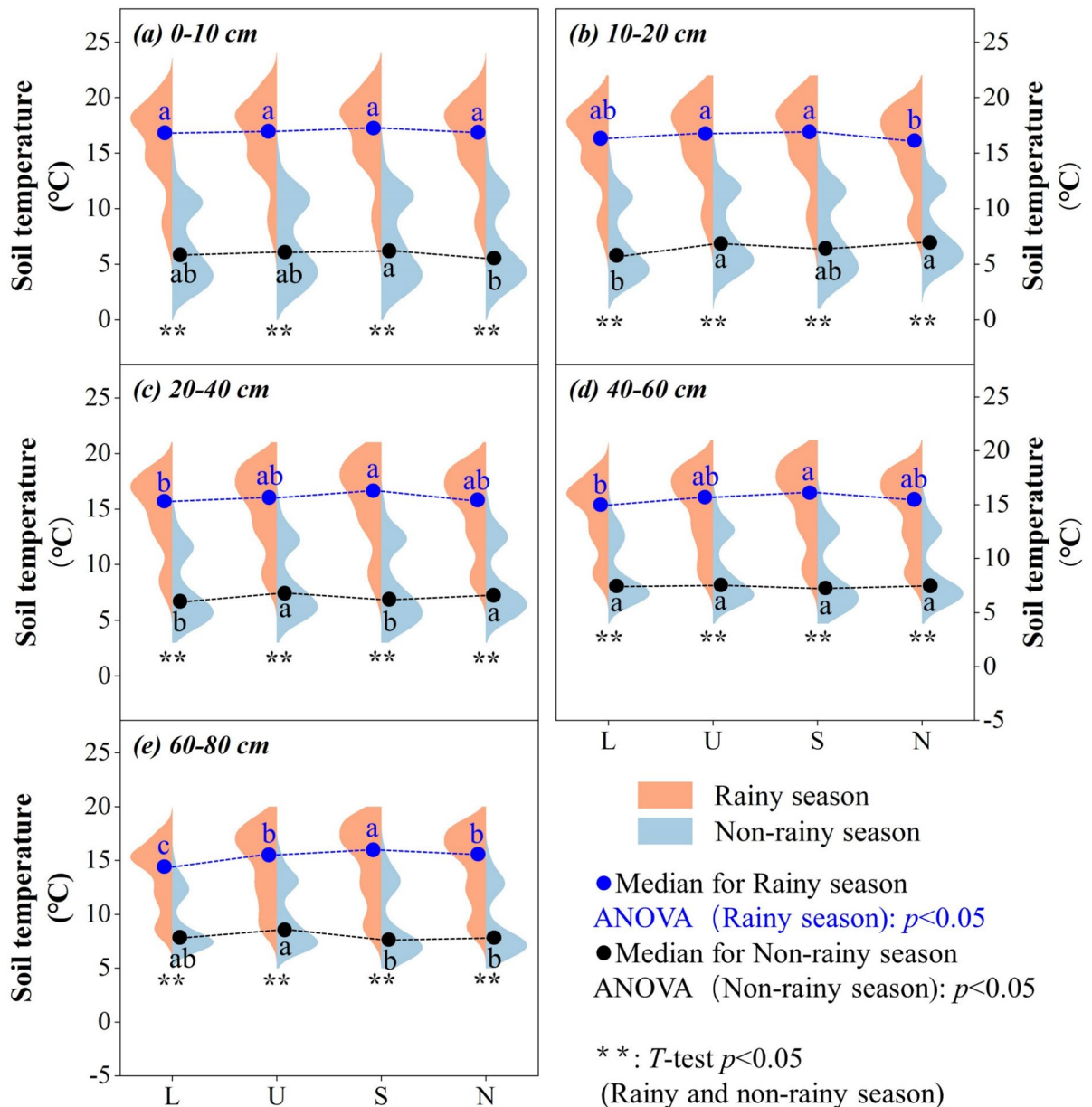


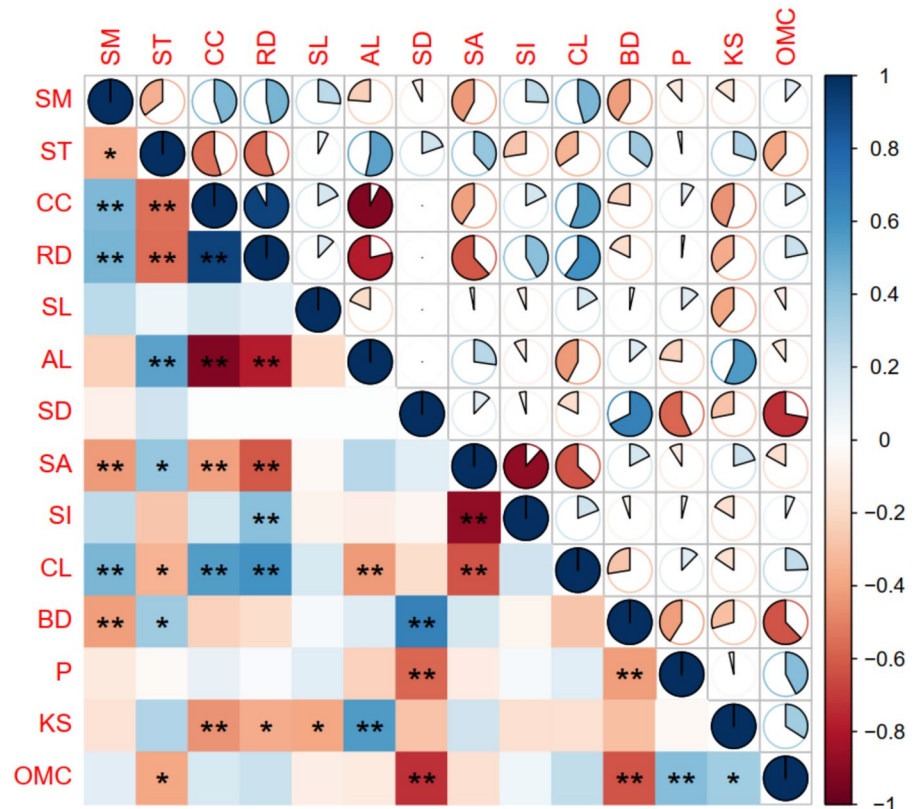
Fig. 5 A comparison of the variability of soil temperature between the rainy and non-rainy seasons in five soil layers. The left half of the violin plot shows the probability distribution of daily soil temperature in rainy season, while the right half depicts soil temperature in non-rainy season. Blue circles depict the median soil temperature in rainy season, while black circles depict non-rainy season. Different letters indicate signif-

icant ($p < 0.05$) differences in average soil temperature among site groups in each season. ** indicates a significant difference ($p < 0.05$) in average soil temperature between two seasons in t -tests. L, lower valley site group; U, upper valley site group; S, south-facing slope site group; N, north-facing slope site group. Each site group includes two monitoring sites

at lower elevations, reflecting the influence of gravitational water movement (Figs. 2 and 3a). In contrast, soil temperature patterns were more influenced by energy inputs, with temporal consistency in thermal

conditions increasing from higher to lower valley locations (Figs. 2 and 3b). This is consistent with previous studies showing that topographic attributes, such as elevation, slope steepness, aspect, and surface

Fig. 6 Relationship between soil micro-climate and vegetation, topography and soil physicochemical properties based on Mantel-test. Color gradients represent Pearson's significance, and figures indicate Pearson's correlation coefficients. SM, soil moisture; ST, soil temperature; CC, mean canopy closure; RD, root depth; SL, slope; AL, altitude; SD, soil depth; CL, clay content; SI, silt content; SA, sand content; BD, bulk density; P, porosity; K_s , saturated hydraulic conductivity; OMC, organic matter content. * and ** indicate the significant differences at the 0.05 and 0.01 levels, respectively

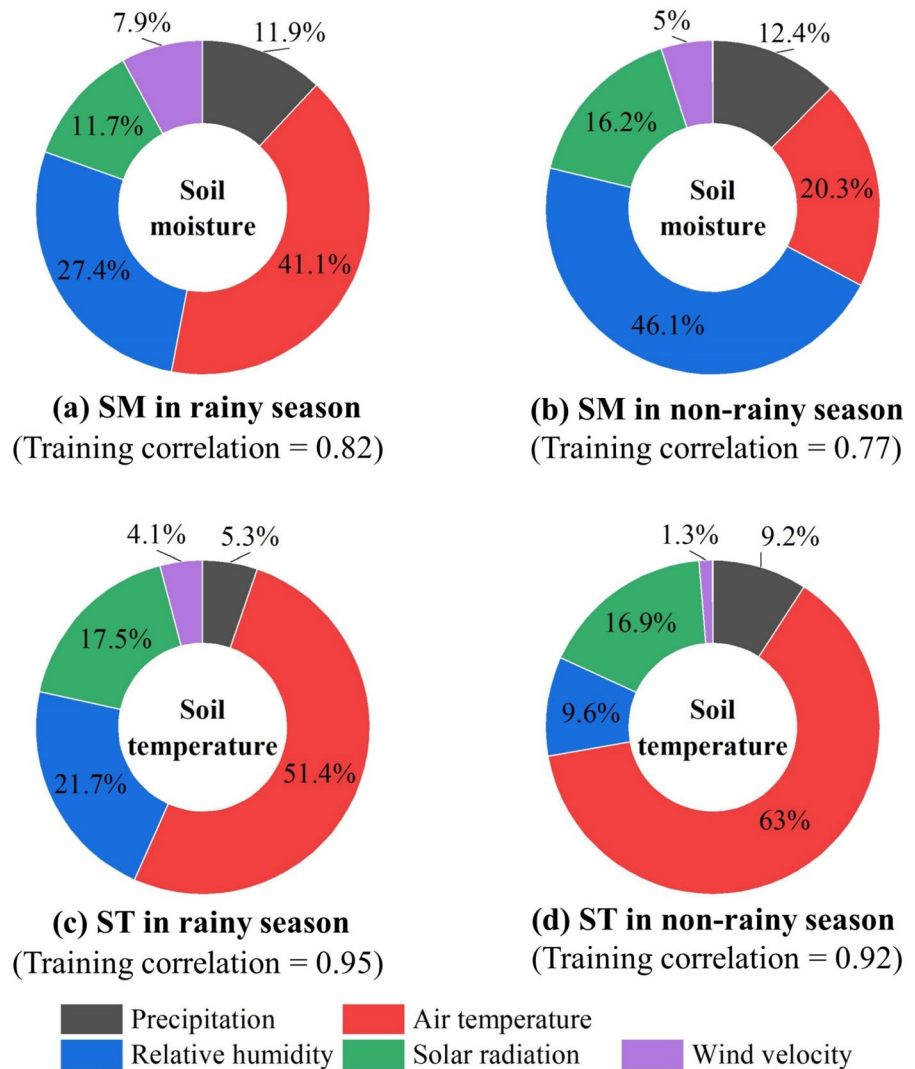


curvature, regulate both soil moisture distribution and solar radiation exposure in vegetated catchments (Fan et al. 2020).

In subtropical mountainous environments where alternating wet and dry periods are common, soil moisture and temperature respond dynamically to meteorological extremes. Areas with steeper gradients displayed more pronounced fluctuations in soil moisture and temperature following rainfall or drought events (Fig. 2; Singh et al. 2021; Souza Wojahn et al. 2023). During dry periods, isolated rainfall events could lead to transient increases in soil moisture, whereas in wetter seasons, the impacts of intense storms on soil moisture would be more sustained (Gao et al. 2020). These hydrological variations interact with thermal processes, jointly influencing how heat is distributed vertically and laterally within the soil matrix (Zhou et al. 2002). A strong link between soil temperature and incoming solar radiation suggests that seasonal energy inputs govern thermal trends, although short-term weather events can cause temporary deviations (You et al. 2013; Cheng et al. 2022).

At broader spatial scales, where atmospheric energy is relatively uniform, terrain-induced differences in air temperature are minimal, though seasonal fluctuations would remain significant (Fig. 5; Jin et al. 2019). However, soil moisture generally increases as elevation decreases. Valley bottoms retained more moisture compared to side slopes, which tended to be drier (Fig. 3). This trend likely results from lateral subsurface water transport that moves moisture downslope, concentrating it in low-lying areas (Guo and Lin 2018). This mechanism has also been documented in humid subtropical hillslopes, where water can migrate laterally between soil layers (Lu et al. 2024). Similar findings have been reported in previous studies of steep catchments with shallow water tables, where terrain strongly would mediate hydrological responses (Lana-Renault et al. 2014; Yu et al. 2018). Interestingly, our data showed that the average soil moisture on south-facing slopes was higher than that on north-facing slopes (Fig. 3). The south-facing slope is exposed to more solar radiation, the heat stored from soil warming facilitates a higher rate of soil moisture extraction through evapotranspiration (Gutiérrez and Vivoni 2013), which

Fig. 7 Contribution of meteorological variables to soil moisture and temperature in rainy season (a, c) and non-rainy season (b, d) determined using BRT analysis. SM, soil moisture; ST, soil temperature



typically results in warmer and drier conditions on south-facing slope (Burnett et al. 2008; Geroy et al. 2011). However, Reid (1973) suggested that in energy-limited systems, where the potential evapotranspiration demand cannot be met, areas favorable for solar radiation would not necessarily lead to drier soils. These findings suggest that the soil micro-climate in the study catchment may not be sufficient to meet the potential evapotranspiration demand (Xu et al. 2025). Furthermore, despite a substantial increase in precipitation during the rainy season, average soil moisture levels did not significantly exceed those observed in the non-rainy season (Fig. 4). This counterintuitive result may

be attributed to increased vegetation activity, including higher transpiration and water uptake during the growing season (Ren et al. 2018; López-Vicente et al. 2023). In addition, the soils in the study area are developed from granite, and numerous fractures and macropores exist within the soil (Lu et al. 2025). Deep percolation or lateral subsurface flow may also reduce the soil's water-holding capacity, which can offset the effect of the increased precipitation (McMillan and Srinivasan 2015; Zhang et al. 2022). In summary, topographic features primarily influence soil moisture dynamics by regulating both vertical infiltration and lateral redistribution processes.

Dominant environmental factors controlling soil micro-climate in the re-vegetation catchment

Although terrain typically exerts a strong influence on the spatial variation of soil micro-climate, its impact can be moderated by vegetation structure and inherent soil characteristics (Geroy et al. 2011; Gutiérrez-Jurado and Vivoni 2013). Even when rainfall and solar radiation are relatively consistent, the interaction among landform, vegetation, and edaphic properties can generate intricate spatial variability in soil moisture and temperature. Moreover, meteorological conditions at the catchment level also play a critical role in shaping soil micro-climatic regimes (Huang et al. 2016; Mao et al. 2024). Understanding how these factors interact is vital for improving our understanding of how soil, vegetation, and atmospheric processes are interconnected in topographically complex ecosystems.

Our analysis revealed strong associations between vegetation traits and soil micro-climate (Fig. 6). Vegetation cover can influence belowground conditions by regulating transpiration and providing shade, thereby influencing both temperature and moisture dynamics (Yu et al. 2022). Canopy interception of rainfall further modifies local soil water input, altering the micro-climate (Brantley et al. 2017; Tanarhte et al. 2024). We found that root depth was positively correlated with SMC, with longer roots typically in downslope areas, explaining the higher moisture levels observed there (Table 1; Fig. 3). Among the various soil traits, bulk density and particle size distribution emerged as key determinants of moisture and temperature regimes (Fig. 6), primarily due to their influence on the soil's ability to retain water (Alaoui et al. 2011; Kang et al. 2022). Since water availability influences heat capacity and thermal conductivity, variations in moisture directly affect how temperature is distributed and changes over time within the soil profile (Zhou et al. 2002; Tao et al. 2017). Overall, both vegetation and soil properties play crucial roles in regulating the soil micro-climate by influencing energy and moisture fluxes.

Although weather variables exert a direct impact on the soil micro-climate, the exact pathways of influence remain complex and not fully understood (Di et al. 2022; Mao et al. 2024). In this study, we found that air temperature and humidity were the dominant drivers of soil moisture variation (Fig. 7a, b). In subtropical mountain

regions characterized by elevated humidity and temperature, these variables critically can affect plant transpiration and water uptake, thereby altering soil water availability (Song et al. 2020). During the rainy season, temperature exerted a stronger influence on soil moisture than humidity, while in the non-rainy season, relative humidity became the more dominant factor (Fig. 7a, b). This pattern suggests that elevated temperatures in the rainy period intensify water loss through vegetation processes, a finding consistent with earlier observations in this area (Xu et al. 2025). Air temperature was also significantly associated with soil temperature across seasons, further emphasizing its influence. In addition, solar radiation emerged as a major factor regulating soil heat content (Fig. 7c, d), indicating that incoming energy is a primary control on soil thermal conditions. Specifically, solar radiation heats the soil surface, influencing soil temperature, while vegetation and soil moisture regulate heat transfer, thereby affecting the rate of soil temperature changes (Jin et al. 2019). This finding aligns with the results of Zhang et al. (2023) in karst regions. By exploring how environmental drivers shape micro-climate conditions at the soil level, this study contributes important insights for improving larger-scale predictions of soil micro-climate behavior.

Limitation, contribution, and outlook

Firstly, this study was conducted in a representative forest watershed in the Three Gorges Reservoir area, where four primary topographic positions were selected for soil micro-climate monitoring (Fig. 1). Each topographic position was equipped with paired monitoring points, one located on the upper slope and the other on the lower slope. Although these sites effectively represent the major topographic regions of a typical fan-shaped drainage system (Liu et al. 2020), the spatial scope of the study remains somewhat limited. Future research could enhance the reliability of the data by expanding the spatial sampling range and incorporating a wider variety of topographic and vegetation settings. Additionally, while this study focused primarily on topographic and soil characteristics, it did not fully consider other important driving factors such as seasonal vegetation dynamics, the redistribution of precipitation by canopy structure, or freeze–thaw processes, all of which are known to influence near-surface microclimates (Mao et al. 2024). To better quantify these interactions, future

studies should integrate targeted field measurement data, such as repeated assessments of the leaf area index to capture phenological changes, and through-fall monitoring to estimate the interception effects under forest cover (Bogena et al. 2018). Furthermore, the use of snow depth markers would provide insights into the spatial variation of snow accumulation and melt. Incorporating this additional evidence into the existing dataset would help provide a more comprehensive analysis of the interactions between vegetation characteristics and soil microclimates, potentially revealing spatial hotspots of biogeochemical activity in forest restoration systems (Andrews et al. 2011).

Secondly, soil moisture and temperature were measured using 5TM capacitive sensors, which are widely used in field hydrological and ecological research due to their ease of operation, low cost, and ability to provide high temporal resolution continuous monitoring. However, the measurements from capacitive sensors are susceptible to a variety of environmental factors, which can potentially introduce biases into the reliability and interpretation of the data (Bosch 2004; Bogena et al. 2010). First, differences in soil texture and bulk density can affect the dielectric constant's response characteristics, leading to systematic biases in sensor outputs for different soil types under the same moisture conditions. Without calibration for specific soil types, this can result in overestimation or underestimation of moisture values (Rosenbaum et al. 2010). Second, temperature variations can alter the dielectric properties of the soil and the sensor's electronic components, thus affecting the stability of the measurements (Bogena et al. 2010; Schwartz et al. 2019). In conditions with significant diurnal temperature fluctuations or seasonal temperature shifts, these thermal effects may be misinterpreted as real moisture dynamics. Future research could improve the overall robustness of soil microclimate monitoring by deploying complementary sensor systems that allow for cross-validation and enhanced calibration accuracy (Fan et al. 2020). It is also important to note that predictions made using the BRT model may be influenced by uncertainties arising from model assumptions, input variables, and sampling design (Sanaei et al. 2019). Future studies should assess the accuracy and reliability of BRT predictions through techniques such as cross-validation,

which will provide further insights into the model's predictive performance and its generalizability across different landscapes. In this study, tenfold method was used to cross-validate the results of the BRT model (Elith et al. 2008; Yuan et al. 2018), showing that the BRT model has good accuracy and reliability.

Thirdly, understanding the spatial distribution of soil moisture and temperature is a crucial guiding factor in vegetation restoration (Jin et al. 2019; Xu et al. 2024). Soil moisture and temperature directly influence plant growth conditions, and different plants have varying requirements for these environmental factors. Therefore, understanding their distribution helps in selecting suitable plant species (García-Ruiz 2010; Zhou et al. 2016). For instance, aquatic or wetland plants are more appropriate for restoration in humid regions, while drought-resistant herbaceous or shrub species are better suited for arid areas. By monitoring soil moisture and temperature, plant selection can be optimized based on specific regional conditions, thereby increasing the success rate of restoration efforts. Additionally, the spatial variations in soil moisture and temperature can guide the design of vegetation restoration layouts. Different topographic features, such as slope and aspect, affect soil moisture retention and temperature changes, so understanding these variations helps in appropriately positioning vegetation to ensure the best growing conditions for plants (Fang et al. 2016; Yue et al. 2020). For example, drought- and heat-tolerant plants should be chosen for high slopes or exposed areas, while wetland plants are more suitable for low-lying or shady regions. Monitoring soil moisture and temperature can also provide guidance for soil management. In areas with excessive moisture, drainage measures can be enhanced, while in arid regions, increasing organic matter can improve soil moisture retention. Furthermore, monitoring these variables helps assess restoration effectiveness (Zhao et al. 2019). As vegetation cover increases, soil conditions change, and by comparing data before and after restoration, adaptive management measures can be taken to ensure ecosystem stability and long-term sustainability. Thus, a comprehensive understanding of the spatial distribution of soil moisture and temperature provides scientific guidance for vegetation restoration, improving both its effectiveness and success rate.

Conclusion

This study investigated a re-vegetated catchment in the mountainous region of the Three Gorges Reservoir Area, aiming to monitor the spatiotemporal distribution of soil micro-climate across different topographic positions and to identify the key environmental controlling factors. The results revealed that soil temperature exhibited relatively consistent seasonal variations across all monitoring sites, with significantly higher values during the rainy season than during the non-rainy season. In contrast, soil moisture was more sensitive to topographic variation, showing higher content in valley areas compared to side-slope positions. These findings demonstrate that continuous monitoring of soil moisture and temperature across heterogeneous topographic settings is an effective means to capture the spatiotemporal variability of soil micro-climate. Moreover, the soil micro-climate is jointly influenced by vegetation characteristics, soil properties, and meteorological variables. Among these, vegetation cover, root depth, soil texture, and air temperature were identified as the primary controlling factors. To advance our understanding of regional soil micro-climate dynamics, future studies should focus on key mechanisms such as air temperature fluctuations, vegetation–soil interactions, and subsurface hydrological processes. Overall, this study provides a valuable foundation for assessing the ecological and hydrological functions in subtropical mountainous regions and supports the formulation of effective vegetation restoration strategies under sustainable land-use management. Looking forward, integrating these insights into predictive ecohydrological models and climate adaptation frameworks could enhance our ability to anticipate soil–vegetation–climate interactions and guide adaptive management practices beyond vegetation restoration.

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Author contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Jiapan Xu, Muxing Liu, Siqi Mei, Qian Yang, Jun Zhang, Jun Yi, Shuting Ding and Yan

Tan. The first draft of the manuscript was written by Jiapan Xu. The academic review and writing supervision were conducted by Muxing Liu and Jun Zhang. All authors read and approved the final manuscript.

Data availability Data will be made available on request.

Declarations

Competing interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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